

Lesson 25 — Design Challenge — Code-Review Gallery

Matches LEGO Teacher Portal lesson: "Culinary Convenience".

Objective (I can...): I can design a well-structured program built from functions and read another team's code to give useful feedback. **CS concept:** Applying functions, parameters, and modular design; peer code review · **Builds on:** Lessons 21-24. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min): Professional AI code is reviewed by peers before it ships — readability and reuse matter as much as "it runs." *First-principles question:* "What would make another engineer trust and reuse your function without rewriting it?"

Materials: LEGO CS & AI kit + device per pair, plan sheets, a short code-review checklist (clear names? one job each? reused? documented?), notebooks. **Model already built** (Functions model) — the 30 min is plan + code + test, small reconfigs only.

60-minute flow - Warm-up (8): Functions review — define/call, parameters, decomposition, refactoring, documentation. (*Theme: Code-review gallery.*) - **Concept + demo (10):** Challenge: build a program that does something interesting using at least two functions, one with a parameter and one reused. Coach models the review checklist on a sample. (*Rotate format: "pair teaches pair."*) - **Build/Code (30):** Pairs plan and code a modular program meeting the function requirements on the built model, documenting each function. *Swap at 15 min.* - **Challenge (7):** Reuse one function from inside a loop or a second function (composition). - **Share + pack-up (5): Code-review gallery** — pairs rotate to another team's device, read one function using the checklist, and write one specific piece of praise and one question in that team's notebook. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator checks structure against the checklist; swap at 15 min. (*Trio rotates plan/code/review.*)

Differentiation: *Support* — one function with a parameter and a provided main outline. *Extension / fast-finisher* — three functions with composition and a parameter passed through. *Reteach* — reuse a Lesson 23 program and add one parameter.

Exit check: Pair delivers a modular program meeting the function requirements and gives one useful, specific review comment to another team. **Progress layer:** Send home the **Unit 5 parent note** ("your student can organize a real program into reusable functions and review code") with the **build photo**. **At-home (optional):** Review a sibling's or friend's plan for anything; give one specific praise and one question, the way we did in class. ~10 min.